

HARIRAMAKRRISHNAN RAMACHANDRAN

+353 89 970 6156 • hariramakrrish@gmail.com • Dublin, Ireland • Stamp 1G — Full-Time Work Eligible

PROFESSIONAL EXPERIENCE

Software Engineer — HCL Technologies, Chennai, India

Sep 2021 – Jan 2025

- Engineered **Python** scripts to automate data extraction and transformation for agricultural emissions reporting, reducing manual processing time by approximately **40%**.
- Developed and validated **greenhouse gas emission models** for diverse agricultural systems, translating complex scientific principles into structured logic for platform integration.
- Collaborated with **VSAg Consultants software developers** to implement, test, and maintain emission calculation models within the VSAg platform, ensuring scientific accuracy.
- Researched and integrated **IPCC methodologies** and national GHG inventory frameworks into model logic, supporting adherence to evolving carbon accounting standards for agriculture.
- Supported the development of new **VSAg modules** by contributing to the design and implementation of emission models for new regions and commodity use cases.
- Authored detailed documentation on model methodologies and scientific underpinnings, enhancing the credibility of the VSAg platform for clients and certifying bodies.

SKILLS

Agricultural Science & Modelling – Greenhouse Gas Emission Models, Agricultural Systems, IPCC Methodologies, Carbon Accounting Standards, Agronomic Science, Climate Action

Data Analysis & Programming – Python, R, SQL, Data Modelling, Quantitative Modelling, Spreadsheets

Software Development & Platforms – Java Spring Boot, REST APIs, AWS, Docker, Linux, Git, Jenkins

BI & Data Visualisation – Power BI, Tableau, DAX, Power Query

Enterprise Systems & Support – ERP Integration, Production Support, Incident Response, Change Management

PROJECTS

Agricultural Emissions Modelling Framework (HCL Technologies)

- Designed and built a modular **Python** framework for simulating greenhouse gas emissions from various agricultural inputs (e.g., fertilizer application, livestock feed).
- Integrated scientific literature data on emission factors and coefficients, enabling the model to adapt to different crop types and farming practices.
- Developed unit tests for model components, achieving over **90% code coverage** and validating output against established benchmarks for tillage and livestock systems.

VSAg Platform Data Integration (MSc Applied Project)

- Developed **SQL** queries and scripts to extract, transform, and load agricultural dataset samples (e.g., soil types, weather patterns) into a PostgreSQL database for analysis.
- Created interactive **Power BI** dashboards visualizing key agricultural metrics and potential emission sources, facilitating stakeholder understanding of data patterns.
- Applied statistical analysis using **R** to identify correlations between farming practices and modelled GHG outputs for a hypothetical dairy farm scenario.

EDUCATION

M.Sc. Data Analytics — National College of Ireland, Dublin

Feb 2026

B.E. Computer Science — SNS College of Technology, Coimbatore, India

2021

CERTIFICATIONS

- IELTS Academic: Overall Band Score **7 out of 9**, issued by IELTS Official, Test Date: 28/AUG/2023
- **Full Stack: Java Spring Boot** and Angular (Great Learning)
- Certificate of Completion in Web Development from **University of California**, Davis (Coursera)
- Certificate of Completion in Python programming from **University of Michigan** (Coursera)